



Article submitted to journal

Subject Areas:

complex systems, statistical physics,
evolution

Keywords:

language change, statistical physics,
phase ordering, spatial inference,
migration, accommodation, replicator
dynamics

Author for correspondence:

Corresponding author

e-mail: james.burridge@gmail.com

Statistical physics of language change inferred from time evolving maps

James Burridge¹, Bert Vaux²

¹ School of Mathematics and Physics, University of Portsmouth

² Department of Theoretical & Applied Linguistics, University of Cambridge

Language change may be studied by tracking linguistic variant frequencies through space and time. Observed frequencies may be viewed as realisations of a statistical field. We first set out an efficient cross validated method for inferring the *empirical* field — the variant frequency distribution estimated directly from data — in our case, high resolution surveys from 20th century USA. We then derive a model of the statistical field, assuming that individuals diffuse locally, periodically migrate, and copy one another based on a form of replicator dynamics involving both bias and linguistic accommodation. We infer model parameters by comparing changes in model and empirical fields. Our inferences demonstrate that spatial linguistic patterns can be rapidly destroyed by migration but that accommodation can create them, protect them, and in combination with diffusion, produce partially predictable pattern dynamics. The results provide the strongest evidence yet for surface tension driven phase ordering in human dialects. We quantify the model's forecasting ability up to a time horizon of twenty five years using the Kullback-Leibler divergence between forecasts and observations.

1. Introduction

Languages typically emerge from interactions between many individual speakers [1]. Because the majority of linguistic exchanges are face-to-face [2, 3], language patterns evolve differently in different places and amongst different social groups. The study of the resulting spatial and social variations in language is called *dialectology* [4]. *Statistical physics* aims to explain the behaviour of physical systems in terms of the properties of their constituent particles. A mathematical object of central importance in statistical physics is the *statistical field* [5], which describes a localised average of the states of the particles which make up the system. The analogy between systems of many physical particles and social systems of many people (or animals) has been widely exploited by statistical physicists [6]. Variations within languages can be analysed using *linguistic variables* [4]. These are linguistic features which can be realised in two or more different forms, known as *variants*. Starting from a population of speakers in a geographical area, we define the "linguistic state field" to be the vector of relative fractions of speakers near to each position who use each variant.

We wish to formulate state field models which explain how collective linguistic patterns emerge from interactions between speakers. Some steps in this direction have already been taken [7–16]. For example, deterministic state field models have been derived, in which speakers preferentially use the variants which they hear more often [17–19]. These models predict the formation of single variant domains whose boundaries — *isoglosses* — evolve by curvature driven dynamics similar to physical *phase ordering* [20]. In the linguistic case, however, population gradients induce forces on interfaces [17, 18]. The predictions of these early models have been tested in a rudimentary way by comparing their equilibria with dialect maps produced by linguists. The results in England [17] and Italy [18], while visually appealing, stopped short of directly inferring parameters from data, or rigorously testing predictive power.

Parameter inference procedures for non spatial stochastic linguistic variant models are well advanced [21, 22]. Such models often exploit the analogy between linguistic and genetic variants [22–26]. A recently developed statistical field model [27] exploits the same analogy, leading to a family of coupled Wright-Fisher diffusion processes. Estimating the parameters of spatial models and testing their predictions is more challenging than in the non-spatial case. First, data is harder to obtain and process. There are non-spatial corpora [28] that track millions of words, but high resolution spatial datasets are smaller and rarer. One reason is that simultaneously collecting spatial and linguistic information from speakers requires their active participation, usually in the form of a survey. Modern online surveys rely on gamification methods or an element of luck in order to become popular. Particularly successful examples are the English Dialect App [29] which collected over 5×10^4 responses, and the Cambridge Online Survey of World Englishes (the dataset we use) which contains close to 10^5 responses [30]. Inference procedures are also harder to devise for spatial models. Some progress has been made using quasi-likelihoods [31], or by tuning parameters to interpolate between two snapshots of spatial variant distributions [19].

While *metaphorical* language models are quite common, spatial models which can be fitted to real observations are rather rare [14–16, 32]. The combination of statistical inference with statistical physics style models of individual linguistic variables is in its infancy. Although a statistical field model now exists [27] with parameters that can be inferred from large scale survey data, it lacks the flexibility to capture realistic movement processes, and is limited to binary linguistic variants.

In this paper we formulate a statistical field model which incorporates realistic long range migration and arbitrary numbers of variants — both of which are essential in order to capture real variables and their dynamics. The model combines diffusion (localised and migratory) with variant selection based on a form of stochastic replicator dynamics [33]. We infer the parameters of this model by first inferring *empirical* statistical fields from data, then inferring model parameters that most closely replicate the dynamics of the empirical fields. Our empirical field inference method avoids over-fitting by using Gaussian process priors [34] and provides the most detailed reconstruction of 20th century American linguistic variables currently available. Our inferred state field model provides the strongest evidence yet for the role of phase ordering in human language via linguistic "accommodation" — speakers adapting their variants to those of their interlocutors. Without accommodation, migration rapidly destroys linguistic variation. We characterise the ability of our model to forecast future change.

2. Inference of empirical fields (time evolving maps)

We begin by presenting our procedure for inferring the time evolution of state fields from data.

(a) Details of survey data

The Cambridge Online Survey of World Englishes (COSWE) [30] contains $\approx 10^5$ responses within mainland USA. The survey poses questions such as *what is your generic casual or informal term for a sweetened carbonated beverage?* Popular responses to this question were *soda*, *pop* and *coke*. These are variants of the sweetened carbonated beverage variable. Most survey respondents reported both their birth date and the geographical coordinates best approximating the location where they acquired their linguistic features (typically the region where they spent most of their childhood). The survey has been running since 2007 and remains active, with birth dates of the majority of respondents (93%) lying in the interval [1950, 2000]. As such it represents a detailed snapshot of early 21st century language use across the post-adolescent age distribution.

In order to infer historical state field evolution we use the *principle of apparent time* [4, 35]. According to this, the linguistic state of post-adolescent speakers approximates the state of the language community to which they belonged during their adolescence. Although the principle has limitations, notably that adult language is not fully stable, comparison of real and apparent time observations support its assumptions [1, 35–37]. We will treat birth year as our time variable. The linguistic state of speakers at *apparent time* t (the state of all speakers

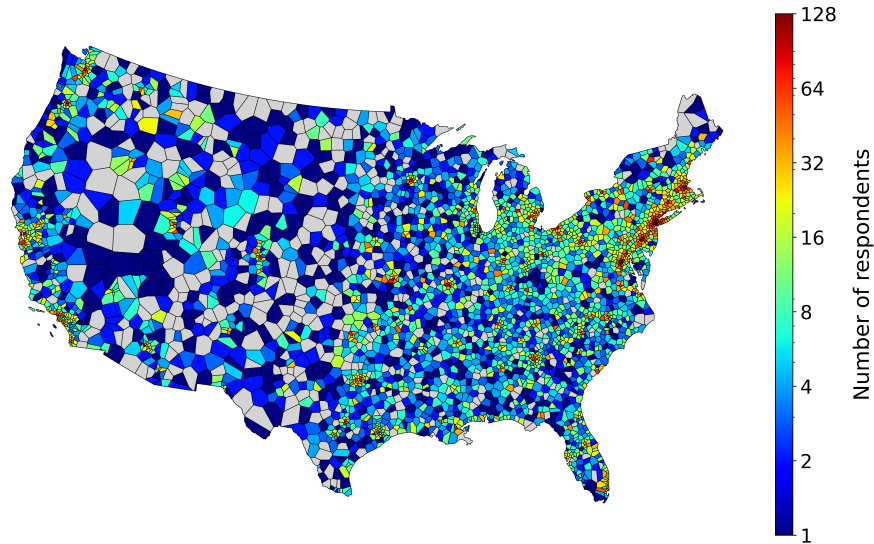


Figure 1. Respondent counts for the athletic shoes variable over the period [1950–2000]. Grey cells contain no data. Spatial cells are a Voronoi tessellation of mainland USA with seeds given by the centroids of 4×10^3 ZIP code clusters obtained via k-means.

born in year t) should be interpreted as a sample from the language community as a whole taken during the *real time* interval $[t, t + a]$ where a is the typical age at which a speaker's language system stabilises. We will consider the apparent time interval [1950, 2000], within which our data is concentrated.

In addition to a small number of common variants, survey questions often also yielded a number of much rarer forms. To simplify our analysis we treat these as a background fluctuation field which we do not explicitly model. In a small number of cases we further simplify the data by grouping similar variants (see Figure 2). We let K be the number of variants remaining after exclusion and grouping. After defining the variants for a question, the response of the i th speaker to this question consists of three variables: their birth year t_i , their coordinates $\mathbf{r}_i = (r_{i1}, r_{i2})$ and their variant $v_i \in \{1, \dots, K\}$. Where speakers specified more than one variant we sampled v_i uniformly at random from this set so each response may be interpreted as a single exemplar 'utterance' by the respondent.

(b) Spatial discretisation

Our models are defined in discrete space. To obtain a discretization which reflects the heterogeneous distribution of people [38] we applied the k-means algorithm [39] to all mainland USA zip code coordinates weighted by their populations as of 2020, to produce $N = 4 \times 10^3$ clusters. The centroids of each cluster were used to seed a Voronoi tessellation (Figure 1) defining the spatial cells of our model. The mean cell population is 82×10^3 with 10th and 90th percentile cell populations 12×10^3 and 207×10^3 .

The heterogeneous population distribution is mirrored by the spatial distribution of survey responses (Figure 1). There are a substantial number of cells which have no data at all. Data sparsity must be borne in mind when interpreting maps of inferred variant

distributions (empirical fields). In sparse areas these will be interpolated from nearby population centres, and represent very small population sizes.

(c) Bayesian inference of empirical linguistic state fields

We wish to estimate the spatial-temporal evolution of linguistic variant frequencies. For a variable with K variants, we let $x_k(\mathbf{r}_i, t) \equiv x_{ik}(t)$ be the relative frequency with which variant k is used within the cell with centroid $\mathbf{r}_i$ at apparent time t . We then define the frequency vector

$$\mathbf{x}(\mathbf{r}_i, t) \equiv \mathbf{x}_i(t) = (x_{i1}(t), \dots, x_{iK}(t))^T.$$

The sum of relative frequencies is by definition one, so frequency vectors define a time evolving statistical field on the simplex Δ^{K-1} . We call $\mathbf{x}(\mathbf{r}, t)$ the *state field* for the variable. To estimate the time evolution of this field given the survey data, we introduce a latent (unobservable) statistical field,

$$\mathbf{F}(\mathbf{r}_i, t) = \mathbf{F}_i(t) = (F_1(\mathbf{r}_i, t), \dots, F_K(\mathbf{r}_i, t))^T$$

which exists in the same space-time domain as the state field, but belongs to $\mathbb{R}^K$ rather than Δ^{K-1} . To condense notation we write $F_k(\mathbf{r}_i, t) = F_{ik}(t)$. The state field is related to the latent field via the softmax function

$$x_{ik}(t) = \text{softmax}(\mathbf{F}_i(t))_k = \frac{e^{F_{ik}(t)}}{\sum_{j=1}^K e^{F_{ij}(t)}}. \quad (2.1)$$

The fact that the latent field is defined on $\mathbb{R}^K$ makes it straightforward to specify a *prior*, encoding assumptions about its ability to fluctuate in space and time [34]. We use a spatial-temporal Gaussian process [34] with four hyperparameters controlling the time-scale (τ) and length-scale (σ) of fluctuations — both via RBF kernels [40] — the sensitivity (α) of length scales to population density, and the overall magnitude of fluctuations (κ). The full definition of the prior is given in the supplemental material [41].

We estimate the state field by first calculating the maximum a-posteriori (MAP) form of the latent field, then applying transformation (2.1). We write the sequence of birth years (apparent times) in our period of interest as a vector $\mathbf{t} = (t_1, \dots, t_T)$. Let F be the tensor with components $F_{ijk} = F_{ik}(t_j)$, and y_{ijk} be the number of respondents in cell i with birth year t_j using variant k , and let Y be the tensor with components y_{ijk} . The probability of observing Y given F (the likelihood) is then

$$p(Y|F) = \prod_{i=1}^N \prod_{j=1}^T n_{ij}! \prod_{k=1}^K \frac{\text{softmax}(\mathbf{F}_i(t_j))_k^{y_{ijk}}}{y_{ijk}!}.$$

Writing the prior density of the latent fields as $p(F)$ then the posterior density satisfies $p(F|Y) \propto p(Y|F)p(F)$ and we estimate the latent fields as

$$\hat{F} = \arg \max_F p(Y|F)p(F).$$

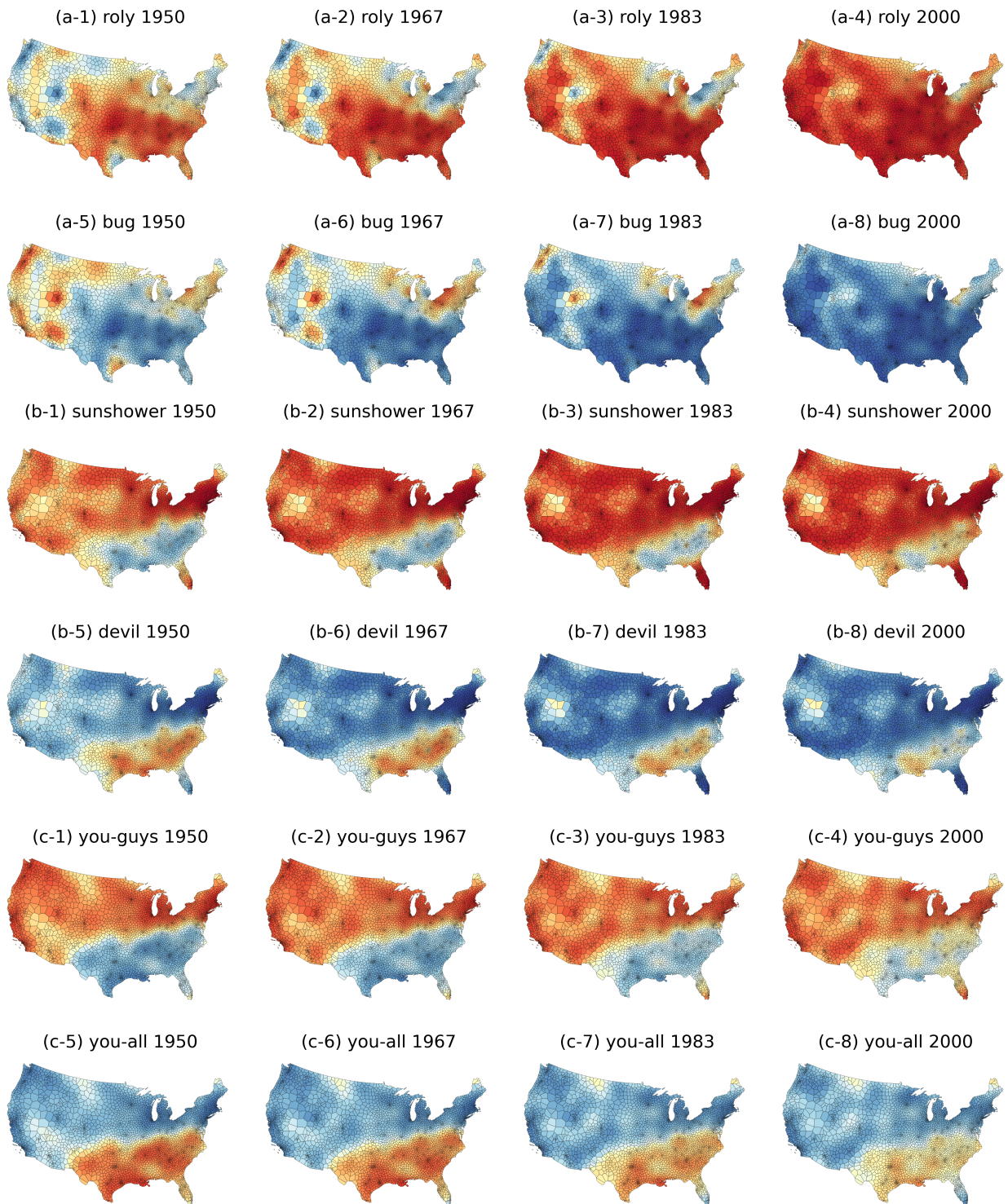


Figure 2. Maps (a-1)–(a-8) show estimated state fields for the variable "What do you call the little gray (or black or brown) creature (that looks like an insect but is actually a crustacean) that rolls up into a ball when you touch it?". Modelled variants: *roly* = "roly poly" and *bug* = {"pill bug", "potato bug"}. "Sow bug" (2%) was excluded, as were respondents with no unique term for this creature (10%). Maps (b-1)–(b-8) show fields for "What do you call the kind of rain that falls while the sun is shining?". Modelled variants: *sunshower* and *devil* = "the devil is beating his wife". "Liquid sun" (1%) was excluded, as were respondents with no variant for this question (48%). Maps (c-1)–(c-8) show fields for the question "What word(s) do you use in casual speech to address a group of two or more people?". Modelled variants *you guys* = {"you guys", "you"} and *you-all* = {"y'all", "you all"}. Excluded variants were "all of you" (6%) and "you people" (2%).

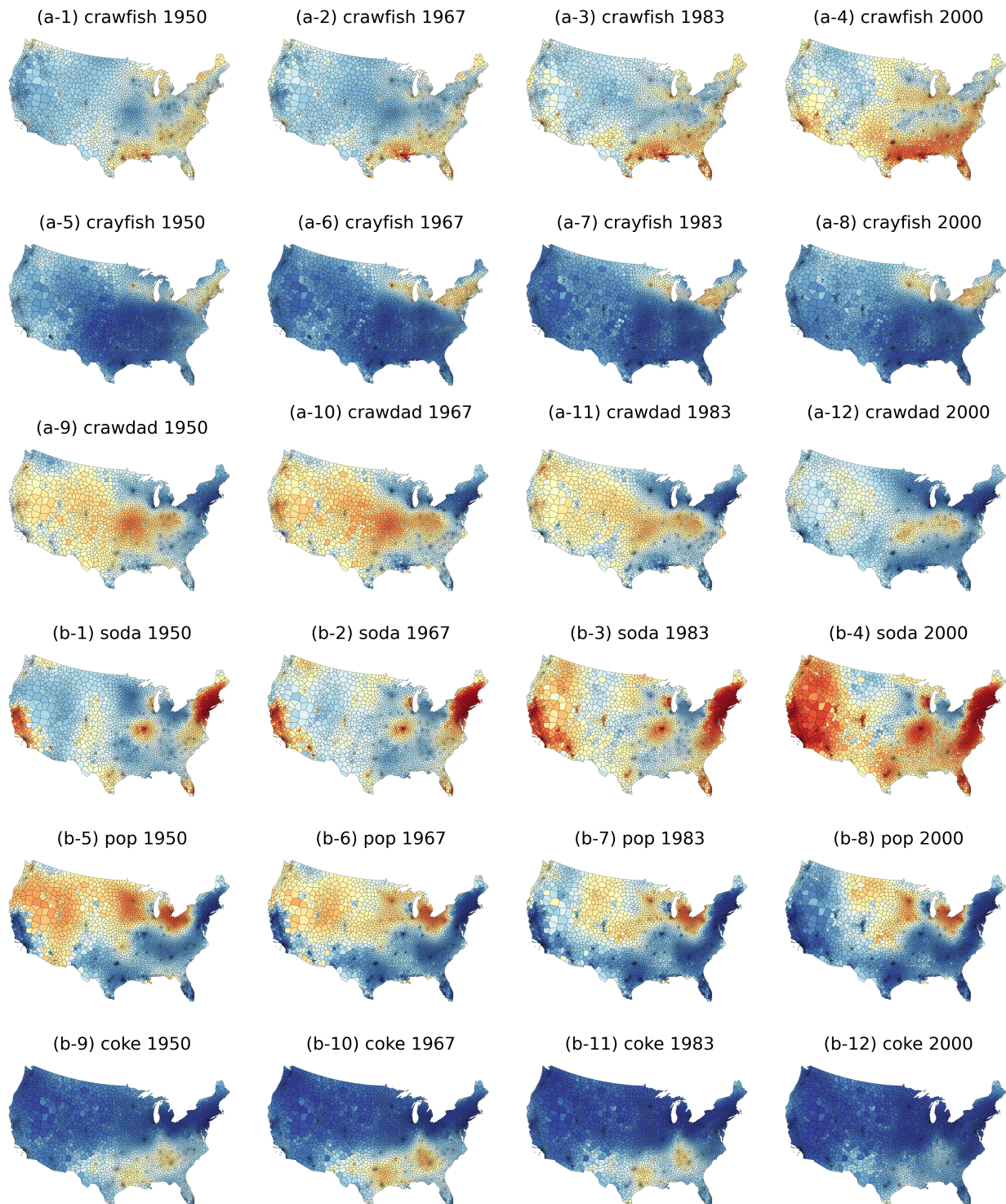
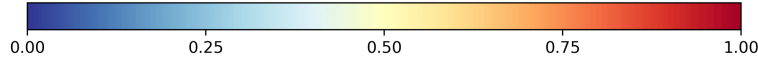


Figure 3. Maps (a-1)–(a-12) show estimated state fields for the variable "What do you call the kind of crustacean that looks like a tiny lobster and lives in lakes and streams?". Modelled variants: *crawfish*, *crayfish*, *crawdad*. Excluded variants ("lobster", "crab", "prawn") have frequencies $\leq 2\%$. Maps (b-1)–(b-12) show estimated state fields for the variable "What is your generic casual or informal term for a sweetened carbonated beverage? (As in: We have milk, beer, apple juice, and four kinds of ...)". Modelled variants: *soda*, *pop*, *coke*. Excluded variants were "soft drink" (7%), "soda-pop" (2%), "fountain drink" (1%). Respondents with no generic term were also excluded.

The optimal hyperparameter vector $\psi = (\tau, \sigma, \alpha, \kappa)$ is selected by first performing a random 80-20 train-test split of the survey data, then maximising (with respect to ψ) the log probability of the test data according to field estimates obtained from the training data. Results for the six variables studied may be found in Figures 2, 3, and 6. We map frequencies to colours using the following scale



The same scale is used in all state field maps throughout the paper.

3. Statistical physics of the state field

We now derive a model of the state field from a model of individual speakers and their interactions.

(a) Diffusion

Each speaker carries within them a set of linguistic variants. The variants carried by a speaker can be transported around physically if the speaker migrates from one location to another, or they can be transmitted from one speaker to another via copying.

It is convenient to think of cells as containing a fixed number of residence sites, each of which contains one speaker. If each speaker within the cell uses a single variant at a given time, then the linguistic state of the speaker in site s of cell i may be represented as a probability vector $\mathbf{V}_{is}(t) \in \{e_1, \dots, e_K\}$ where e_k is the standard unit vector in the k th coordinate direction. If the speaker in site s of cell i is using variant k at time t then $\mathbf{V}_{is}(t) = e_k$. The state field in cell i is related to the states of individual speakers as follows

$$\mathbf{X}_i(t) = \frac{1}{P_i} \sum_{s=1}^{P_i} \mathbf{V}_{is},$$

where P_i is the cell population. The state field is capitalised to indicate that it is a random vector.

One way in which the state field can change is by speakers migrating between cells. The distance between cells located at $\mathbf{r}_i$ and $\mathbf{r}_j$ is defined $d_{ij} = \|\mathbf{r}_i - \mathbf{r}_j\|$. In time δt we write the probability that an individual migrates from $\mathbf{r}_i$ to $\mathbf{r}_j$, where $i \neq j$, as $w_{ij}\delta t$ where w_{ij} is the single speaker transition rate. We assume w_{ij} obeys the following gravity-type model [42, 43]

$$w_{ij} = cP_i^{\alpha-1}P_j^{\alpha}(d_0 + d_{ij})^{-\gamma_0-\gamma_1 \log(d_0+d_{ij})},$$

with parameters $c > 0$, $\alpha \in [0, 1]$, $d_0 > 1$, $\gamma_0 > 0$, $\gamma_1 \in \mathbb{R}$. We ignore within-cell migration since it does not change the state field. Hence we set $w_{ii} = 0$. The total migration rate per speaker out of cell i is $w_i = \sum_j w_{ij}$, with transition rates between cells depending on their separations and their respective populations. The flow rate of migrants from $\mathbf{r}_i \rightarrow \mathbf{r}_j$ is $f_{ij} = P_i w_{ij}$.

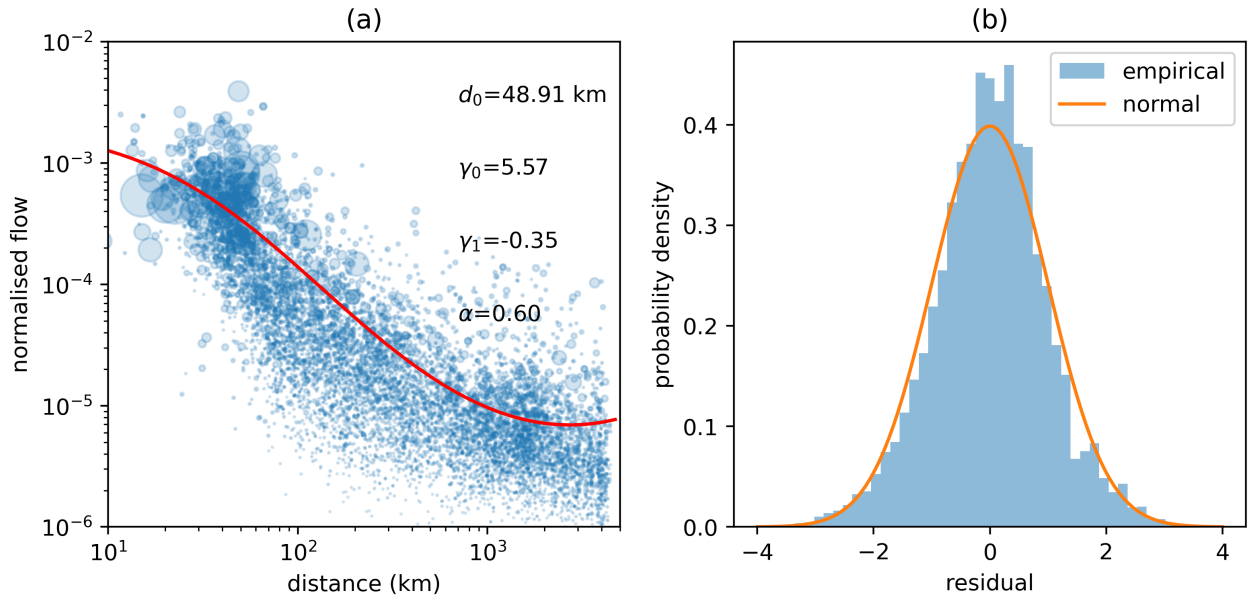


Figure 4. (a) Scatter plot of normalised 2011 county-county migration flow rates $f_{ij}/(P_i P_j)$ between cells with point sizes proportional to absolute flow rates. Red curve shows inferred flow rate model, with parameter values annotated on the plot. (b) Histogram of residuals between the observed and modelled log normalised flows. Orange curve is a normal density fitted to the histogram.

We estimate parameters by minimising the squared residuals (weighted by flows) between empirical and model estimates of *normalised* flow rates $y_{ij} = f_{ij}/(P_i P_j)$, where empirical flows are extracted from county-county migration data [44, 45]. Details of the fitting procedure are given in the supplemental material [41], and the results are displayed in Figure 4. We choose the constant c so that the model matches historical migration rates. The average rate per person is given by

$$\lambda = \frac{c}{P} \sum_{i,j} f_{ij}$$

where P is total population. During the period [1950,2000] (excluding years 1972–75 and 1977–80 for which the CPS 1-year mobility question was not asked) this rate remained stable [46] with mean $\bar{\lambda} = 6.3\%$ (per person per year probability of moving county) and standard deviation 0.4% . We select c so that $\lambda = \bar{\lambda}$.

To model cell-to-cell variant diffusion via copying we assume that in time δt , each speaker in cell i will copy the variant of speaker selected uniformly at random from cell j (possibly equal to i) with probability $l_{ij} \delta t$, where

$$l_{ij} = \frac{P_j \exp\left(-\frac{d_{ij}^2}{2R^2}\right)}{\sum_k P_k \exp\left(-\frac{d_{ik}^2}{2R^2}\right)}.$$

According to this definition, speakers are more likely to copy variants from more densely populated cells which are near to them. The length scale R is set equal to a small multiple of the typical cell radius $\bar{R}_{\text{cell}} = 25\text{km}$ (we set $R = 100\text{km}$).

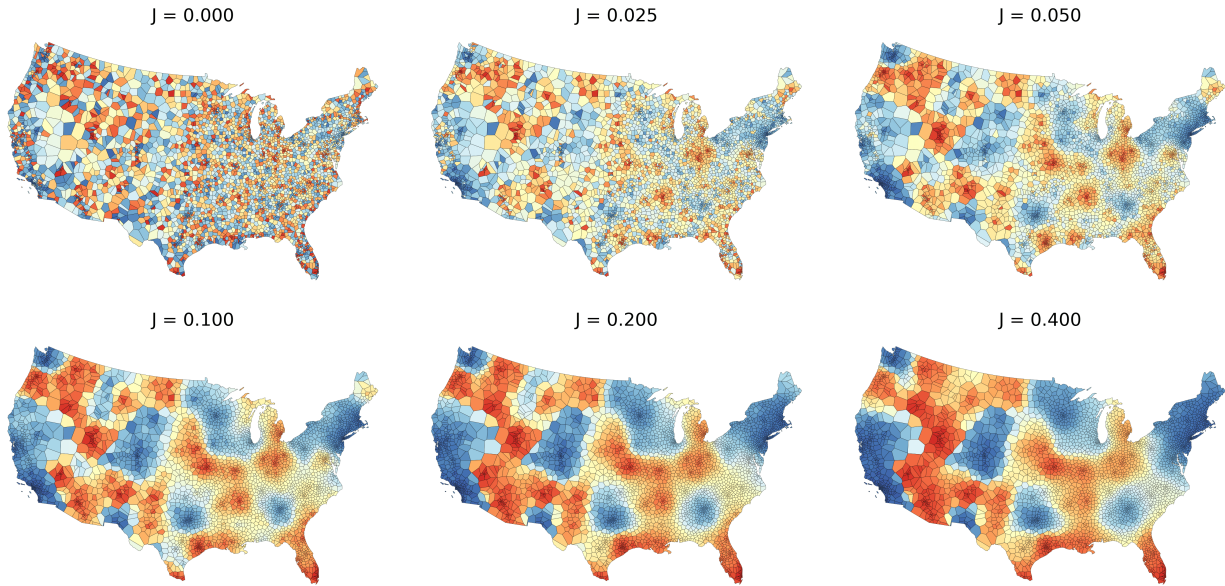


Figure 5. End results of 50 years of evolution in two variant system beginning with a "primordial" state where each cell initialised with a state field selected uniformly at random from the simplex. Each map corresponds to a different value of the copying rate J , with $R = 100\text{km}$. Selection dynamics is unbiased but has an accommodation factor $\beta = 0.2$ (see section 3(b)) which is typical of the factor inferred from survey data (see section 4).

The strength of variant diffusion due to this copying process is determined (in the limit of small cells) by the derived parameter JR^2 [27]. Hence the choice of R serves only to set the typical scale of J . It may be shown [41] that the expected change in the state field due to migration and copying in time δt is

$$\mathbb{E}_t(\delta \mathbf{X}_i(t)) = \sum_j (w_{ij} + J l_{ij} - (w_i + J) \delta_{ij}) \mathbf{X}_j(t) \delta t \quad (3.1)$$

where $\delta \mathbf{X}_i(t) = \mathbf{X}_i(t + \delta t) - \mathbf{X}_i(t)$ and $\mathbb{E}_t$ denotes expectation conditional on the state of the system at time t . From this we see that increasing the rate of local copying is equivalent to increasing physical migration between nearby cells.

The copying rate, J , plays an important role in determining the shape and dynamics of *interfaces* between geographical regions within which one variant dominates. Because its effect is felt predominantly at interfaces, it cannot easily be inferred by optimising model fit at all cells simultaneously. We therefore set its value so that model interfaces reproduce empirical patterns of variant use. Figure 5 shows state field evolution from a randomised initial condition, in each case using model parameters typical of those inferred from data (this includes parameters for both migration and selection—described in section 3(b) below). In each example we use a different value of J . When $J = 0$, corresponding to the absence of local copying, we see that the system remains highly disordered. As we increase J we begin to see single-variant domains appear with boundaries that become increasingly smooth. This behaviour is seen in physical systems which phase-separate, and is known as phase ordering [20]. Applying Occam's razor, we set $J = 0.1$ which is just large enough so that

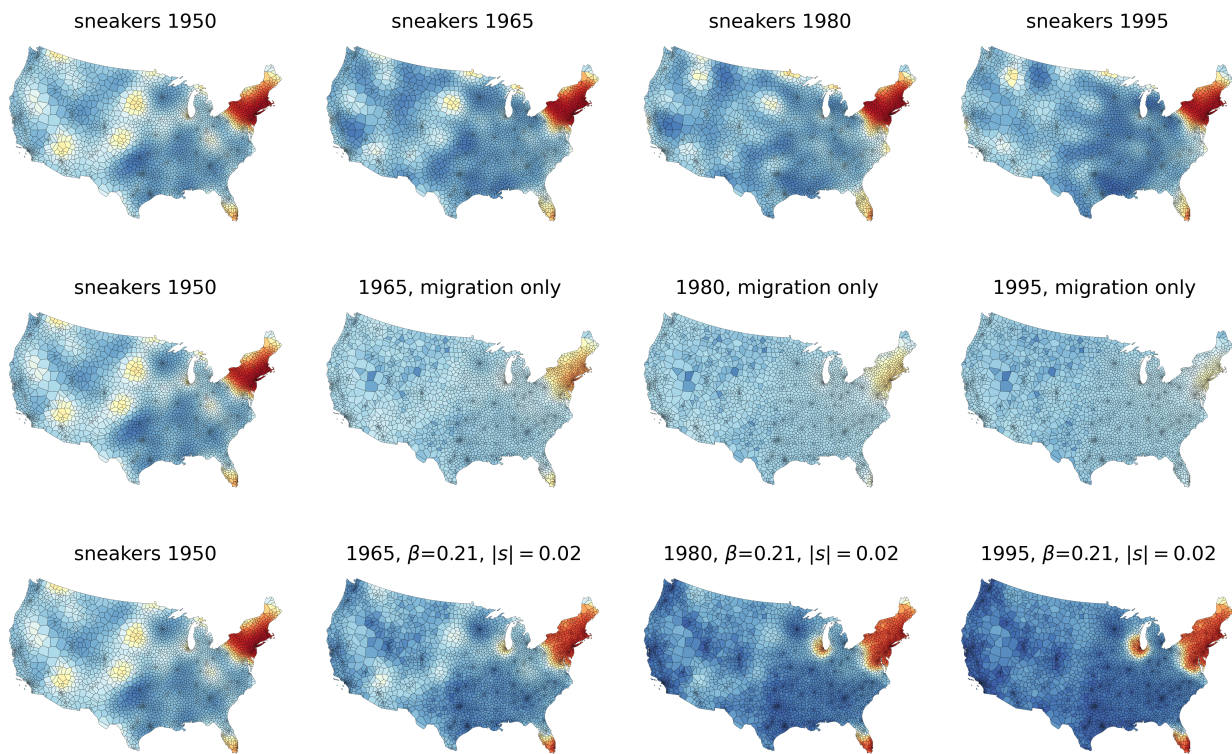


Figure 6. Top row shows the 1950-1995 MAP evolution of the *sneakers* variant derived from the variable "What is your general, informal term for the rubber-soled shoes worn in gym class, for athletic activities, etc.?". Modelled variants were *tennis* = "tennis shoes" and *sneakers* = {"sneakers", "sneaks"}. Excluded variants were "gym shoes" (8%), "running shoes" (7%), "shoes" (5%) with others $\leq 3\%$. Second row shows evolution of model, initialised with 1950 MAP fields, using *only* diffusion. Final row shows evolution of model with same initial condition, but with constant (spatially invariant) bias field and accommodation factor inferred from data (see section 4).

our model is capable of spontaneously generating interfaces of the form seen empirically in Figures 2 and 3.

(b) Selection

When defining cell-to-cell variant copying we assumed that all variants were equally attractive. In the language of genetics, all variants have the same "fitness" [22, 47]. The equal fitness assumption fails to capture some important aspects of variant transmission. First, individuals may preferentially select variants which are more common locally. Such *frequency dependent selection* is known in linguistics as "accommodation" [48]. It models the advantage gained by using the same speech forms as those with whom you are talking [49]. Second, *biases* may exist toward different variants. These might be linguistic (ease of articulation, perceptual distinctiveness, learnability, consistency with wider grammatical or phonological patterns) or social, signifying identity, status, prestige or stigmatization [1, 50]. Third, our model is necessarily a simplified characterisation of a complex interacting system of speakers. Spatially varying fitness then serves as an effective parameter, absorbing the aggregate influence of unmodelled processes.

To illustrate the importance of accommodation, Figure 6 shows the distribution of the "sneakers" variant between 1950 and 1995. The main alternative variant is "tennis shoes", which dominates outside of the Northeast. The "sneakers" region has remained stable with a relatively sharp interface or isogloss running from Washington to Cleveland. The second row in Figure 6 shows our prediction for what would have happened to this pattern if the only evolutionary forces at play were diffusive — it would have largely disappeared. The final row shows what happens if we include accommodation and a small bias (the modelling details are given below). Here the sneakers region survives because the effect of speakers accommodating to local norms acts fast enough to undo the diluting effect of incomers who use different variants.

To model variant selection we introduce a *variant fitness vector* [33] in each cell $\mathbf{q}_i = (q_{i1}, \dots, q_{iK})^T$ where $q_{ik} > 0$ is the fitness of variant k . We assume that in time δt a speaker in site s of cell i who currently uses variant j , will transition to variant k with probability

$$\Pr(\mathbf{V}_{is}(t + \delta t) = \mathbf{e}_k | \mathbf{V}_{is}(t) = \mathbf{e}_j) = \begin{cases} q_{ik} X_{ik} \delta t & \text{if } j \neq k \\ 1 - (\bar{q}_i - q_{ij} X_{ij}) \delta t & \text{if } j = k \end{cases}$$

where $\bar{q}_i = \mathbf{q}_i^T \mathbf{X}_i$ is the mean fitness of all the variants in the cell. When all variants have the same fitness, then according to this model, speakers update as if they are copying the variant of another speaker selected uniformly at random from within their cell. According to our fitness-based selection model, the change in state of an individual during time δt , given by $\delta \mathbf{V}_{is}(t) = \mathbf{V}_{is}(t + \delta t) - \mathbf{V}_{is}(t)$, has expectation

$$\mathbb{E}_t(\delta \mathbf{V}_{is}(t)) = (\mathbf{q}_i \circ \mathbf{X}_i(t) - \bar{q}_i \mathbf{V}_{is}(t)) \delta t$$

where $\circ$ denotes the Hadamard (element-wise) product. The expected state vector increment is then

$$\mathbb{E}_t(\delta \mathbf{X}_i(t)) = (\mathbf{q}_i \circ \mathbf{X}_i(t) - \bar{q}_i \mathbf{X}_i(t)) \delta t. \quad (3.2)$$

In the limit of large numbers of sites per cell evolution becomes deterministic, and (3.2) may be recognised as the replicator equation [33], which is invariant under the transformation

$$\mathbf{q}'_i(\mathbf{x}) = \mathbf{q}_i(\mathbf{x}) + c(\mathbf{x}), \quad (3.3)$$

where $c(\mathbf{x})$ is an arbitrary scalar function. In order for the stochastic transition rates to remain positive the fitness vector must remain in $\mathbb{R}_{>0}^K$, so not every scalar shift yields a well defined stochastic model. Shifts that produce a well defined model change the rate at which speakers switch variants and therefore influence the stochastic component of the dynamics, with larger fitness values leading to faster updating. We are interested only in estimating parameters which determine the expected changes, so we will allow inferred fitness values to be negative on the understanding that these must be shifted to yield a well defined stochastic model.

The fitness vector combines intrinsic bias and accommodation as follows

$$\mathbf{q}_i(\mathbf{x}) = \mathbf{s}_i + \beta \mathbf{x}$$

where $\mathbf{s}_i = (s_{i1}, \dots, s_{iK})^T$ is the *bias vector*, and β is the *accommodation factor*. Larger β values make it more advantageous to select variants that are more frequent in the cell, further enhancing their frequency.

The collection of bias vectors define a spatially varying vector bias field which can be inferred from data. Allowing this field to vary independently between cells will lead to overfitting. To reduce the effective dimension of the field we define the bias matrix $\mathbf{S} \in \mathbb{R}^{N \times K}$ with elements $[S]_{ij} = s_{ij}$ in terms of a lower dimensional latent bias matrix $\Psi \in \mathbb{R}^{Q \times K}$ with components ψ_{ij} . We define the relationship between $\mathbf{S}$ and Ψ via a *lifting matrix* $\mathbf{A} \in \mathbb{R}^{N \times Q}$ as follows

$$\mathbf{S} = \mathbf{A}\Psi. \quad (3.4)$$

The lifting matrix is constructed [41] to induce spatial correlations with typical minimum length scale η . Combining diffusion and selection, we can write our model as a system of stochastic differential equations

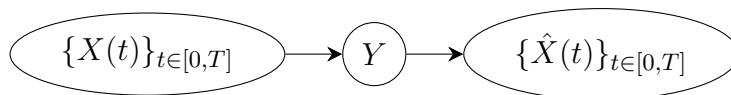
$$d\mathbf{X}_i = \left(\mathbf{q}_i \circ \mathbf{X}_i(t) - \bar{q}_i \mathbf{X}_i(t) + \sum_j (w_{ij} + J l_{ij} - (w_i + J l_i) \delta_{ij}) \mathbf{X}_j(t) \right) dt + d\boldsymbol{\xi}_i(t) \quad (3.5)$$

$$= \mathbf{b}_i(\mathbf{X}; \boldsymbol{\theta}) dt + d\boldsymbol{\xi}_i. \quad (3.6)$$

Here $\mathbf{X} \in \mathbb{R}^{N \times K}$ is the matrix of state field values, $\mathbf{b}_i$ is the drift function, $\boldsymbol{\theta}$ is the parameter vector of the model, and $d\boldsymbol{\xi}_i(t)$ is an approximately Gaussian noise increment [41]. Our focus is on inferring the parameters of the drift vector, rather than on statistics of the noise, which we set to zero when performing predictive simulations.

4. Inference of state field physics

The behaviour of the state field model is determined by its parameters. We infer these by minimising residuals between drift increments, and increments in MAP (empirical) state fields. We assume the model is well specified in the sense that state field trajectories represent samples from our state field model for some $\boldsymbol{\theta}$. Let $\hat{\mathbf{X}}_i(t)$ be our MAP estimate of the state field at time t and let $\hat{\mathbf{X}}(t) \in \mathbb{R}^{N \times K}$ be the matrix of state field values in all cells. The MAP field follows a stochastic process whose realised values depend on the survey data, and therefore also on the historical (but not directly observable) state field trajectories. The dependence of the MAP process on the true historical process $\mathbf{X}(t)$ over the observation interval $[0, T]$ is summarised by the graph



where Y is the tensor of variant counts. The central assumption behind our fitting procedure is that increments in the MAP state field may be expressed as

$$\delta \hat{\mathbf{X}}_i(t) = \mathbf{b}_i(\hat{\mathbf{X}}(t); \boldsymbol{\theta}) \delta t + \delta \boldsymbol{\xi}_i(t), \quad (4.1)$$

where the residual increment $\delta\mathcal{E}_i$ includes contributions from the stochastic component of the state field process and sampling noise in the MAP estimate, and satisfies the moment condition $\mathbb{E}(\delta\mathcal{E}_i(t)) \approx 0$ where the expectation is taken over state field and sampling process. We do not expect noise increments to be independent. Since each MAP field noise increment is realised only once, we replace the moment condition with an empirical analogue — the mean residual over all cells and time increments

$$\frac{1}{NT} \sum_{i=1}^N \sum_{j=0}^{T-1} \left(\hat{\mathbf{X}}_i(j\delta t) - \mathbf{b}_i(\hat{X}(j\delta t); \boldsymbol{\theta}) \right) \approx 0$$

where T is the number of time increments required to cover our time interval of interest. We find $\boldsymbol{\theta}$ such that this condition is satisfied as closely as possible by minimising the regularised mean squared residual

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \left(\frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T \|\delta \hat{\mathbf{X}}_i - \mathbf{b}_i(\hat{X}; \boldsymbol{\theta}) \delta t\|^2 + \frac{\gamma}{QK} \sum_{j=1}^Q \sum_{k=1}^K \psi_{jk}^2 \right)$$

where $\gamma \approx 10^{-4}$ is a small penalty which regularises the latent bias field components and imposes the soft constraint that their mean is zero. The MAP fields may be understood as being derived from the state fields via a combination of smoothing and noise (the extent of smoothing is determined by the priors). In equation (4.1) we have assumed that the state and MAP fields have the same drift. This is justified provided that drift function varies approximately linearly over the time and distance scales of the smoothing.

(a) Migration, accommodation and interfaces

Consider the athletic shoe variable (Figure 6) for which we inferred a single accommodation factor and spatial constant bias field for the full data interval [1950, 2000]. This variable is unusual in that its distribution has remained nearly constant. As discussed in section 3(b), this stability is made possible by the accommodation factor, without which migration would rapidly destroy the spatial variations present in 1950.

To understand how accommodation induces and maintains interfaces, we consider the continuum limit of our model along a single spatial direction parameterised by spatial coordinate z , assuming uniform population density and $K = 2$ variants, which we will call A and B. We assume the state field is invariant along the direction orthogonal to z . To simplify notation we write the frequency, X_1 , of variant A at displacement z and time t as $x(z, t)$. We write the bias vector as $\mathbf{s} = (s, -s)$, the fraction of incoming migrants using variant A as $\bar{x}(z, t)$ and the individual migration rate as λ . We then have

$$\frac{\partial x}{\partial t} = D \frac{\partial^2 x}{\partial z^2} + \lambda(\bar{x} - x) + 2sx(1 - x) + \beta x(1 - x)(2x - 1) \quad (4.2)$$

where $D = JR^2/2$ is an effective diffusion coefficient [27].

In the absence of bias and accommodation ($s = \beta = 0$), the diffusion and migration terms will equalise variant usage throughout the system, leaving it in a spatially uniform

state. In linguistics this process is referred to as "levelling" [2]. Now consider a spatially homogeneous system in which bias and accommodation are present. Spatial homogeneity implies that $\bar{x} = x$, yielding dynamics

$$\dot{x} = sx(1 - x) + \beta x(1 - x)(2x - 1).$$

Provided that $|s| < \beta$ then this admits two stable steady states at $x = \pm 1$ and an unstable equilibrium at $x^* = (\beta - 2s)/(2\beta)$. The system will converge either to $x = 1$ or $x = 0$ depending on whether $x(0) > x^*$ or $x(0) < x^*$.

When $s = 0$, then (4.2) also admits a steady state solution consisting of (one or more) interfaces between regions where different variants dominate. Consider a single interface centred on $z = 0$. Near $z = 0$, provided there are no other interfaces within migration range, then the average state of incoming migrators will be $\bar{x} \approx 1/2$. Defining $x = \frac{1}{2} + u$ then steady state solutions of (4.2) satisfy

$$Du'' + au - 2\beta u^3 = 0,$$

where $a = \beta/2 - \lambda$. Provided $\beta > 2\lambda$ this equation admits the solution

$$u(z) = \sqrt{a/(2\beta)} \tanh(\sqrt{a/(2D)}z),$$

corresponding to the stable interface

$$x(z) = \frac{1}{2} \left(1 + \sqrt{1 - \frac{2\lambda}{\beta}} \tanh \left(\frac{1}{2} \sqrt{\frac{\beta - 2\lambda}{D}} z \right) \right). \quad (4.3)$$

Introducing bias or more interfaces within migration range will cause the interfaces to move over time. In particular, if A is the majority variant amongst incoming migrators, then the A-B interface will move to expand the A domain unless opposed by a sufficiently large bias toward B speakers.

The above analysis has implications for real dialect patterns. First, the condition $\beta > 2\lambda$ quantifies when spatially heterogeneous linguistic patterns can exist. In our tennis-sneakers example we found that $\beta \approx 0.2$, which is more than twice the empirically measured USA migration rate $\lambda \approx 6\%$. This is consistent with the preservation of the sneakers region provided that its interface with the tennis shoes region does not move inwards due to the effects of bias or migration. Second, equation (4.3) provides an analytical expression for the width of linguistic interfaces (isoglosses)

$$w(D, \beta, \lambda) = \sqrt{\frac{D}{\beta - 2\lambda}}.$$

Using $\beta = 0.2$, $\lambda = 0.06$, $J = 0.1$, $R = 100\text{km}$ we obtain $w \approx 80\text{km}$, about twice the average cell diameter of 44km. Inspection of the inferred tennis-sneakers state fields reveals that the isogloss is two to three cells wide, consistent with this width estimate, and with our estimate of J .

If we generalise our continuum model to arbitrary two dimensional state fields — by replacing ∂_x with the Laplacian ∇^2 — we obtain a linguistic analogue of the time

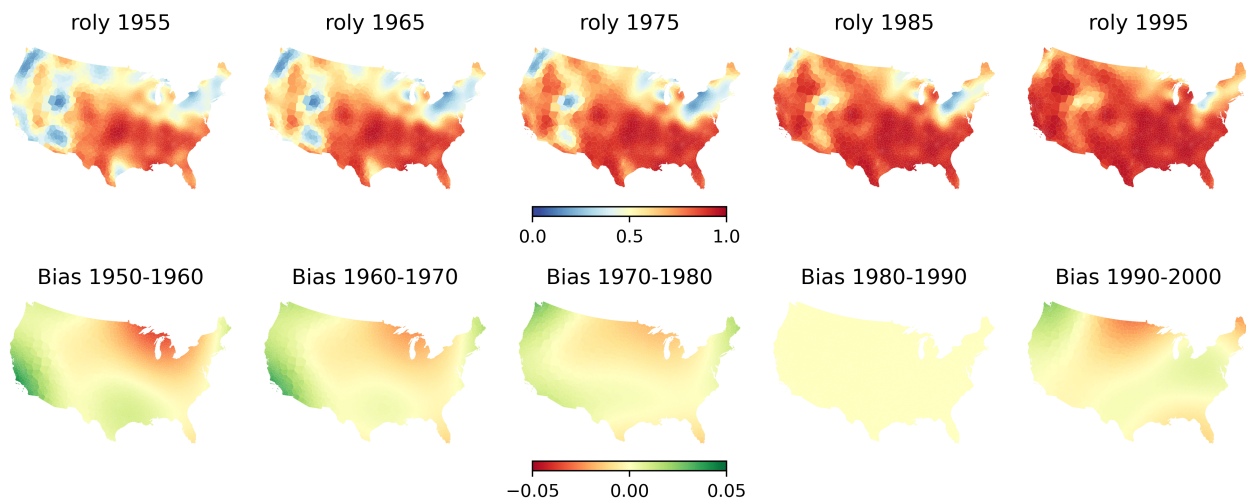


Figure 7. Top row shows mid-decade MAP state fields for the *roly* variant of the variable "What do you call the little gray (or black or brown) creature (that looks like an insect but is actually a crustacean) that rolls up into a ball when you touch it?". Bottom row shows bias fields (length scale $\eta = 1000\text{km}$) for this variant inferred using MAP fields within each decade. Accommodation factors inferred for each decade are $\hat{\beta} = 0.19, 0.20, 0.19, 0.15, 0.10$.

dependent Ginzburg-Landau equation for phase ordering in physical systems [20, 51]. The linguistic analogue of single physical phase domains are single linguistic variant domains, with interfaces driven by a combination of curvature reduction and population density gradients [17]. The population gradient effect causes interface motion to be biased *away* from population centres — down the population density gradient — meaning their linguistic features are to some extent protected from external influence. In the linguistic case accommodation plays the same role as local alignment processes in physical systems. The bias term acts like an external field, biasing interface motion, expanding domains in which the higher fitness variant dominates. The fact that the motion of interfaces evolving to minimise curvature is predictable in physical systems suggests that the evolution of linguistic domains should, to some extent, also be predictable.

(b) Spatial bias field and forecasting

We now consider the model with spatially varying bias field, fitted to variables whose distributions have changed over the study period. Figure 7 shows the evolution of the MAP state field for the term "roly-poly" used to describe a woodlouse (the terrestrial arthropod, Oniscidea). Also shown is the bias field (length scale $\eta = 1000\text{km}$) inferred using MAP fields for each decade from the 1950s to the 1990s. This variant has expanded from the South and southern Midwest into the North and West, eventually becoming dominant throughout the USA. From 1950-1980 the inferred accommodation factor was $\hat{\beta} \approx 0.2$ (matching the tennis-sneakers variable), which is above the threshold required to maintain spatial variation when the average incoming state is an equal balance of variants. In fact the *roly* variant was in the majority in 1950 (see Figure 11), meaning that in most regions the majority of incoming migrants would have been *roly* users. We might expect this to have shifted interfaces in favour of *roly*. In the case of uniform population distribution this would occur, but in Los

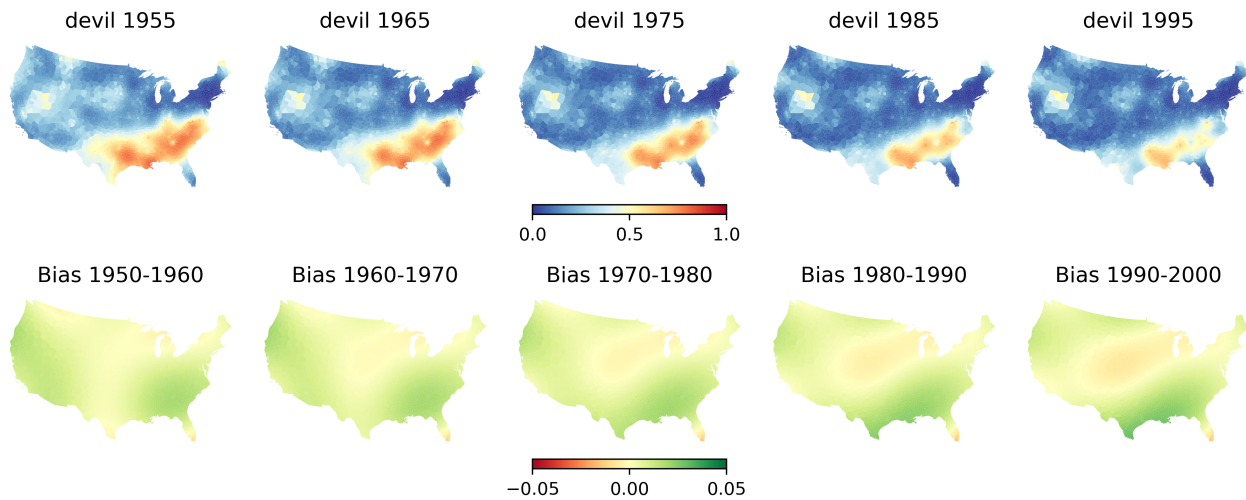


Figure 8. Top row shows mid-decade MAP state fields for the *devil* variant of the variable "What do you call the kind of rain that falls while the sun is shining?". Bottom row shows bias fields (length scale $\eta = 1000\text{km}$) for this variant inferred using MAP fields within each decade. Accommodation factors inferred for each decade are $\hat{\beta} = 0.18, 0.17, 0.16, 0.13, 0.09$.

Angeles (and other cities) a combination of population gradients and geographical distance from other population centres provides a protective effect. To explain the colonisation of Los Angeles by "roly poly" using our model therefore requires a strong positive bias ($\hat{s} \approx 0.04$) in favour of the roly variant in the Los Angeles region, sufficient to "flip" the region away from the *bug* variant. As roly becomes increasingly dominant in the country as a whole, the flow of roly speakers into bug regions will increase, pushing the system toward the pure roly state. Later, a similarly large bias was needed to flip the Pacific Northwest region (Seattle and Portland).

An alternative explanation for the change in Los Angeles is that the population of Los Angeles County grew from 2.7 million in 1940 to 7 million in 1970, requiring unusually high rates of inward migration [52]. This would have resulted in a bias toward the national majority variant. In order to resolve the relative importance of migration and bias, we would need a migration model capable of reproducing the historical population movements more accurately.

A second and somewhat counter-intuitive example of the interaction between bias, accommodation, and migration is the shrinkage of the *devil* variant of the variable "What do you call the kind of rain that falls while the sun is shining?" (Figure 8). In this case we infer a small positive bias in its favour throughout the period of interest, and gradually declining accommodation factor. Spatial variations in bias are smaller than in the previous example. The fact that the devil domain shrinks despite positive bias in its favour may be explained by a combination of incoming speakers whose average state is closer to the *sunshower* variant, and to a decline in the protective effect of accommodation.

As well as providing a simple model-based explanation of observed changes in the state field, the model may also be used as a method of predicting future changes. To assess its ability to make predictions we infer model parameters over a decade long "training period",

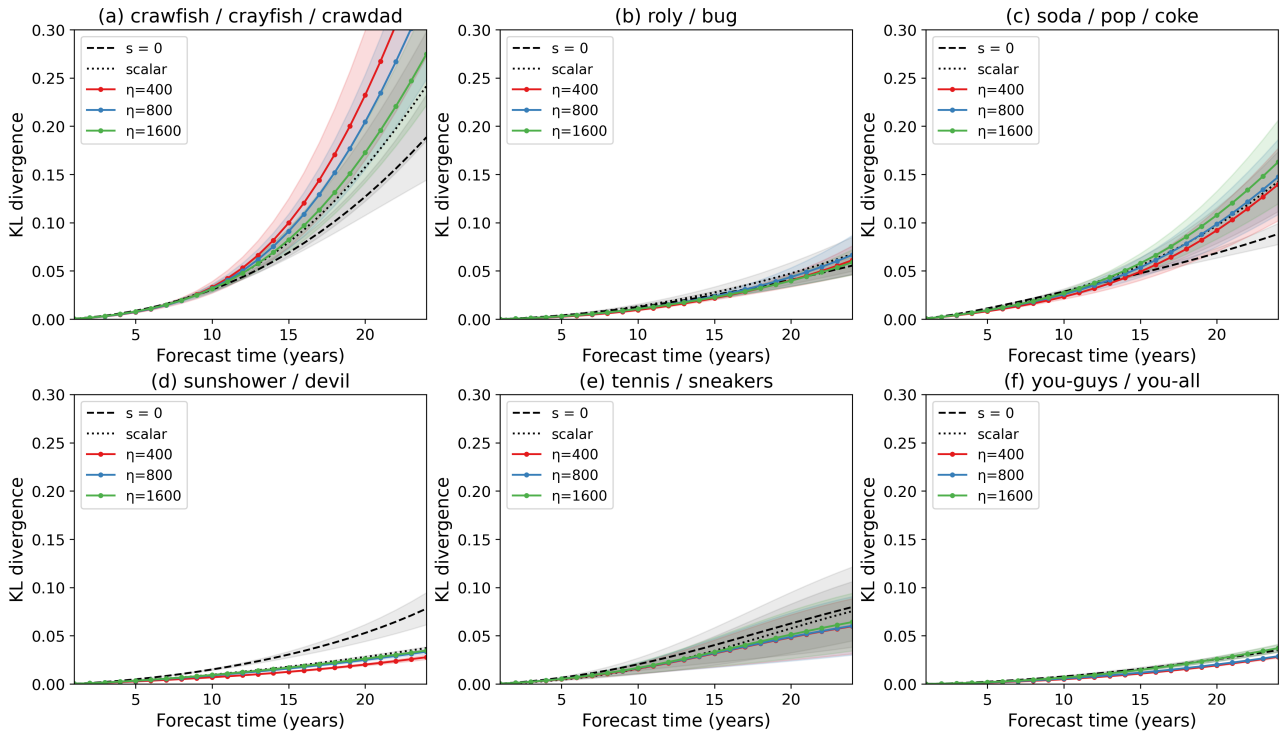


Figure 9. KL divergence between model forecasts empirical fields over a twenty five year forecast window, using parameter estimate from ten year training windows. Bands show ± 1 standard deviation in the KL divergence values over all training windows.

initialise the model with the MAP field at the end of the period, and evolve it forward in time by solving the system of differential equations (3.6) with $d\xi_i = 0$. In each cell we compute the Kullback-Leibler (KL) divergence [53] between the predicted and MAP state fields at future time t

$$D_{\text{KL}}(\hat{\mathbf{x}}_i^{\text{MAP}}(t) \parallel \hat{\mathbf{x}}_i^{\text{pred}}(t)) = \sum_{k=1}^K \hat{x}_{ik}^{\text{MAP}}(t) \log \frac{\hat{x}_{ik}^{\text{MAP}}(t)}{\hat{x}_{ik}^{\text{pred}}(t)}.$$

The predictive performance of the model at time t is then defined to be the average KL divergence over all cells at that time. Repeating the process for a series of overlapping training decades and bias field length scales allows us to estimate the mean and standard deviation of this performance measure over different prediction horizons from different starting points. We also calculate the performance of the constant bias and zero bias models using the same method. Figure 9 shows the relationship between KL divergence and forecast horizon for each of these models. In panels (b) and (f) we see minimal difference in the forecasting ability of different models; in both cases the inferred biases are close to zero (see also Figure 11). In panels (a) and (c) all models perform similarly well initially (ten to fifteen years into the future), after which the more complex models perform worse than the zero bias model. For reference, Figure 10 illustrates 5, 10 and 15 year forward predictions from the model for the *soda*, *pop* and *coke* variants, with $\eta = 2000\text{km}$, with corresponding KL divergences of 0.007 (5 year), 0.02 (10 year) and 0.04 (15 year). In panels (d) and (e) of Figure 9 the situation is reversed — the presence of a bias field improves forecasts from five years onward. Taken together these observations suggest that when forecasting a decade

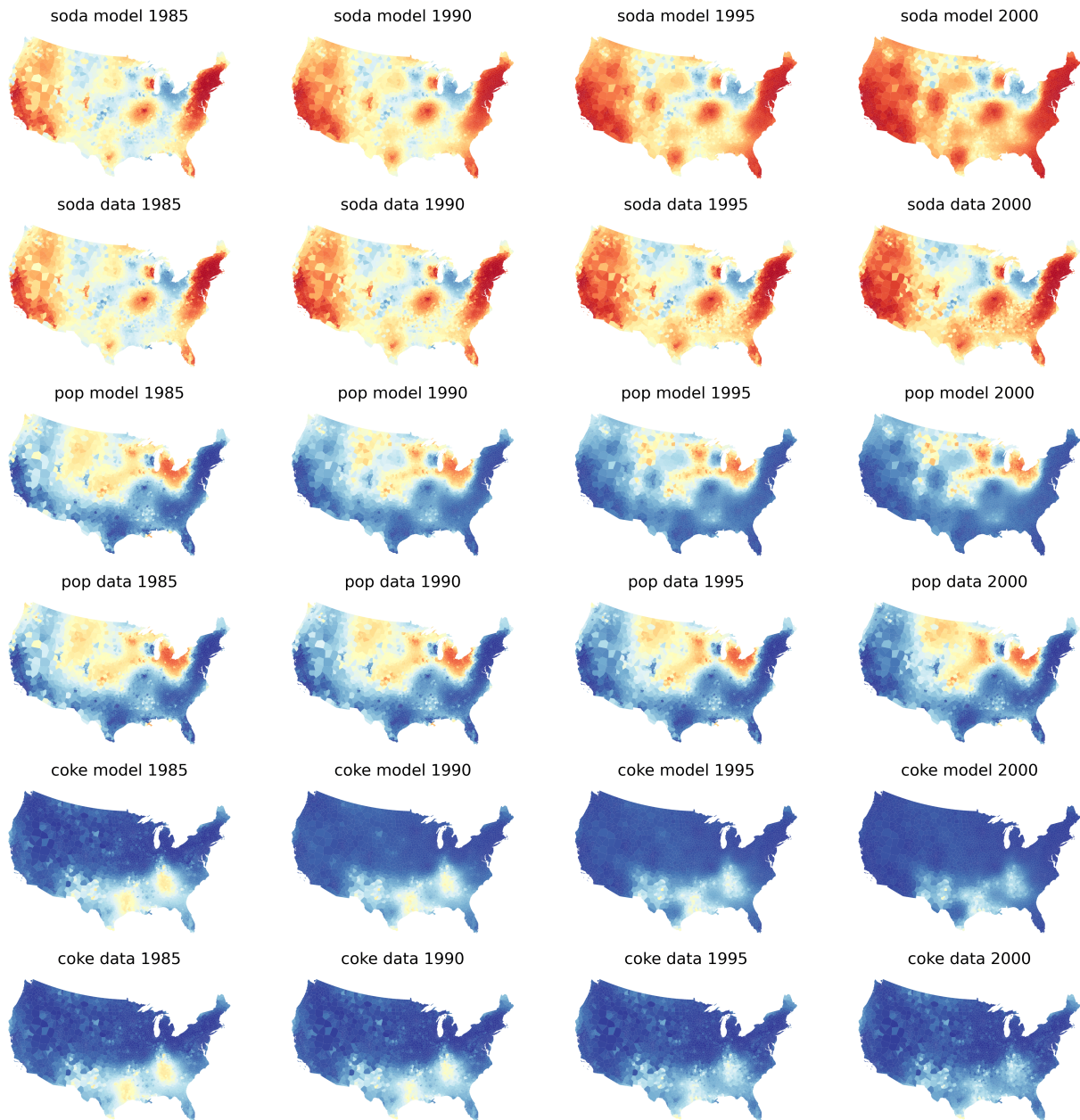


Figure 10. Forward predictions of the *soda*, *pop* and *coke* state fields using model parameters fitted to MAP fields in the training interval [1975,1985], with $\eta = 2000\text{km}$. Inferred accommodation factor $\hat{\beta} = 0.20$.

into the future, bias fields inferred from the recent past improve (or do not worsen) future predictions. However spatial variations in bias inferred from the past do not, in general, improve future predictions.

(c) Time evolution of parameters for all variables

In Figure 11 we have fitted the model with constant spatial bias field ($\eta = \infty$) using a decade-long sliding window of MAP fields from the 1950s to the 1990s. This allows us to see how our model parameters evolve through time. Figure 11 also shows how the proportions of

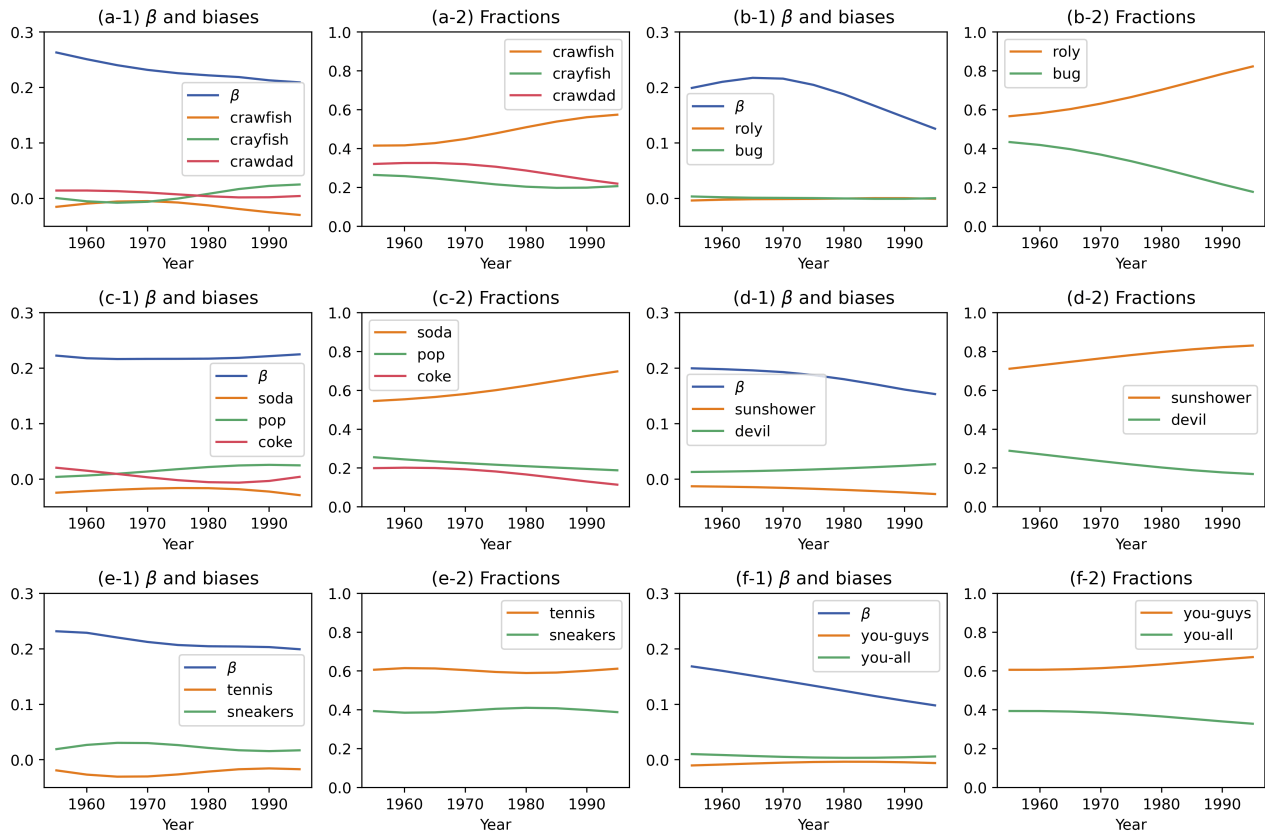


Figure 11. Panels (a-1) to (f-1) show accommodation factors and inferred spatially homogenous biases ($\eta = \infty$) for every variant of each variable. Parameters were inferred using a sliding decade long window of MAP fields, and plotted at decade mid-points. Panels (a-2) to (f-2) show fractions of the population using each variant over the same time period.

each variant in the US mainland as a whole have changed over the same period. Our first observation is that the accommodation factor is remarkably consistent across variants and through time, and comfortably large enough to prevent migration from levelling variation across the country. The fact that the accommodation factor is large enough to play a significant role in dynamics is one of the key findings of our work. While previous studies have hypothesised accommodation (or "conformity") as an important driver of linguistic pattern dynamics [17, 19, 22, 27, 49], statistical evidence for its importance has been difficult to obtain, largely due to the fact that spatial linguistic data is usually only available as a snapshot in time [4], and often of low resolution.

For all but one variable we see a small decline in accommodation over the study period, consistent with a reduction in the distinctiveness of local linguistic variations. A possible explanation for this is that interactions (as opposed to physical migration) have become less localised over time, perhaps due to greater audio-visual media influence. Since this effect is not explicitly contained in our model, it is possible that the inferred accommodation factor is shrinking to compensate.

In four out of the six variables plotted in Figure 11 (panels (b), (c), (d) and (f)) the inferred bias for the variant which has *increased* in proportion over the study period is zero or negative. This counter intuitive result may be explained by considering the combined effect

of migration and accommodation. In each of these examples the growing variant — which we'll call A — begins in the majority. Consequently, in the majority of locations the fraction of A speakers will increase due to accommodation. Further, the majority of migrators will also be A-speakers, exerting an effective bias in favour of this variant. A negative bias inferred for an expanding variant therefore implies that it is growing somewhat slower than would be expected given migration and accommodation alone. Of course this picture is simplified: the geography of real populations can protect minority regions [17, 19], and in regions where variants signify local identity, there will be a positive bias in their favour [54].

5. Linguistic perspective

We now consider the implications of our results for linguistics. The inferred state fields reveal several patterns of significant interest from a dialectological perspective. The poly / pill bug boundary (Figure 2a) largely tracks the traditional boundary between the South and North Midland dialect regions established by Kurath 1949 [55], lending confidence to the state field inference method (section 2): we recover a well-documented dialect boundary from survey data alone, without any prior knowledge of traditional dialect regions. The sunshower / devil beating his wife maps (Figure 2b) are particularly striking, since the devil-based expressions, which are a regionally and socially marked form associated with Southern and African American vernacular traditions, show marked contraction over the study period despite a small inferred positive bias in their favour, suggesting that accommodation and migration are jointly overwhelming what appears to be a socially embedded prestige or identity effect. This is consistent with Labov's distinction between markers (variables subject to social evaluation) and indicators (variables below social awareness): identity-linked forms may retain positive bias yet still recede under demographic pressure [1, 50, 56]. The soda / pop / coke variable (Figure 3b) illustrates a three-way competitive system in which regional boundaries have remained remarkably stable across the study period, with coke confined to the South and pop dominant across the Upper Midwest and Pacific Northwest, a pattern whose persistence through half a century of high migration the model (sections 3 and 4) now quantitatively explains via the accommodation mechanism. The sneakers / tennis shoes isogloss (Figure 6), running roughly from Washington DC to Cleveland, is one of the sharpest lexical boundaries in American English and has long attracted attention precisely because it resists the levelling one would expect given the scale of post-war population movement in the Northeast corridor; the model's demonstration that accommodation alone is sufficient to account for the maintenance of this boundary, without requiring any special social salience or prestige effect, is a significant result.

There are a number of areas where our modelling results are relevant to ongoing debates in linguistics. There has been a long-running debate in sociolinguistics, going back to Trudgill's work on dialect contact and new-dialect formation [2, 49], about whether accommodation suffices to explain the persistence of dialect boundaries, or whether additional factors such as social identity, prestige, and stigma are necessary. Our model

provides the first quantitative demonstration, fitted to real spatial data over decades, that accommodation alone (at empirically inferred rates of $\beta \approx 0.2$) is sufficient to maintain sharp isoglosses against realistic levels of migration. This doesn't rule out social factors, but it establishes a precise quantitative baseline. There has been considerable debate, particularly in British sociolinguistics [2, 57, 58], about whether increased mobility necessarily leads to dialect levelling. Our model gives a rigorous answer: levelling is the outcome only when the migration rate exceeds half the accommodation factor ($\lambda > \beta/2$). This converts what has been a qualitative debate into a testable quantitative condition, and empirically, for 20th century USA, the condition for maintaining variation is comfortably met. Our paper uses the apparent-time construct [35] to infer historical evolution. The fact that the inferred trajectories are coherent, smooth, and consistent with independently-known dialect geography provides, to our knowledge, the most large-scale quantitative validation of the apparent-time assumption yet available.

6. Conclusions

We have made two methodological contributions toward the development of a statistical field theory for language. The goal of this discipline is to explain how coarse grained patterns in language use emerge from interactions between individual speakers. In particular, we want to understand the extent to which the evolution of the state field is predictable, and what kinds of collective behaviour can emerge. To address these questions quantitatively requires modelling, and models must answer to data. Our view is that statistical physics is the ideal tool to develop such models, but that this must be combined with practical methods of inference.

Our first contribution was an efficient method for inferring the (apparent time) historical evolution of state fields. We made use of Gaussian process priors and cross validation in order that our inferred fields provided a maximally plausible model of the true fields. To our knowledge, the resulting maps provide the most detailed examples of the evolution of American linguistic features over the second half of the 20th century. Our second contribution, closely related to the first, was to define a new multi-variant statistical field model which was capable of explaining the observed fields in terms of realistic individual diffusion and selection processes. These processes included migration, local diffusion, linguistic accommodation and spatially varying "biases". We inferred the parameters of our model, excluding stochastic terms, by comparing changes in our historical fields with changes predicted by the model. Our inferences showed that accommodation and migration are powerful creative and destructive forces in linguistic pattern formation, and the pattern evolution is partially predictable.

Data Accessibility. The code and state-field data generated by this study are available from the public GitHub repository https://github.com/james-burridge/COSWE_stat_phys

Authors' Contributions. Burridge developed the mathematical and statistical methodology. Both authors contributed to interpretation and analysis of results.

Funding. The authors are grateful for the Royal Society APEX award APX\R1\241139 which supported this research

References

- ¹W. Labov, *Principles of Linguistic Change, Volume 1: Internal Factors* (Blackwell, Oxford, 1994).
- ²P. Trudgill, *Dialects in Contact* (Blackwell, Oxford, 1986).
- ³J. Milroy and L. Milroy, "Linguistic change, social network and speaker innovation", *Journal of Linguistics* **21**, 339–384 (1985).
- ⁴J. Chambers and P. Trudgill, *Dialectology*, Cambridge Textbooks in Linguistics (Cambridge University Press, 1998).
- ⁵M. Kardar, *Statistical Physics of Fields* (Cambridge University Press, Cambridge, 2007).
- ⁶C. Castellano, S. Fortunato, and V. Loreto, "Statistical physics of social dynamics", *Reviews of Modern Physics* **81**, 591–646 (2009).
- ⁷D. M. Abrams and S. H. Strogatz, "Modelling the dynamics of language death", *Nature* **424**, 900 (2003).
- ⁸M. Patriarca and E. Heinsalu, "Influence of geography on language competition", *Physica A* **388**, 174–186 (2009).
- ⁹A. Kandler, R. Unger, and J. Steele, "Language shift, bilingualism and the future of Britain's Celtic languages", *Philosophical Transactions of the Royal Society B* **365**, 3855–3864 (2010).
- ¹⁰K. Prochazka and G. Vogl, "Quantifying the driving factors for language shift in a bilingual region", *Proceedings of the National Academy of Sciences USA* **114**, 4365–4369 (2017).
- ¹¹M. Mussa Juane, L. F. Seoane, A. P. Muñuzuri, and J. Mira, "Urbanity and the dynamics of language shift in Galicia", *Nature Communications* **10**, 1680 (2019).
- ¹²J. Burridge and T. Blaxter, "Using spatial patterns of English folk speech to infer the universality class of linguistic copying", *Physical Review Research* **2**, 043053 (2020).
- ¹³J. Burridge, T. Blaxter, and B. Vaux, "Evolutionary paths of language", *EPL (Europhysics Letters)* **128**, 28003 (2019).
- ¹⁴H. Kauhanen, D. Gopal, T. Galla, and R. Bermúdez-Otero, "Geospatial distributions reflect temperatures of linguistic features", *Science Advances* **7**, eabe6540 (2021).
- ¹⁵T. Takahashi and Y. Ihara, "Quantifying the spatial pattern of dialect words spreading from a central population", *Journal of The Royal Society Interface* **17**, 20200335 (2020).
- ¹⁶I. Lazzizzera, "Founding a mathematical diffusion model in linguistics: The case study of German syntactic features in the North-Eastern Italian dialects", *The European Physical Journal Plus* **138**, 1–18 (2023).
- ¹⁷J. Burridge, "Spatial Evolution of Human Dialects", *Phys. Rev. X* **7**, 031008 (2017).
- ¹⁸J. Burridge, "Unifying models of dialect spread and extinction using surface tension dynamics", *Royal Society Open Science* **5**, 171446 (2018).
- ¹⁹J. Burridge and T. Blaxter, "Inferring the drivers of language change using spatial models", *Journal of Physics: Complexity* **2**, 035018 (2021).
- ²⁰A. J. Bray, "Theory of phase-ordering kinetics", *Advances in Physics* **51**, 481–587 (2002).
- ²¹J. G. Montero and R. A. Blythe, "Self-contained Beta-with-Spikes approximation for inference under a Wright–Fisher model", *Genetics* **225**, iyad092 (2023).
- ²²R. A. Blythe and W. Croft, "S-curves and the mechanisms of propagation in language change", *Language* **88**, 269–304 (2012).
- ²³G. J. Baxter, R. A. Blythe, W. B. Croft, and A. J. McKane, "Utterance selection model of language change.", *Phys. Rev. E* **73**, 046118 (2006).
- ²⁴R. A. Blythe, "Neutral Evolution in Language", *Advances in Complex Systems* **15**, 1–17 (2012).
- ²⁵S. Wright, "Evolution in Mendelian Populations.", *Genetics* **16** 2, 97–159 (1931).
- ²⁶R. A. Fisher, *The Genetical Theory of Natural Selection* (Clarendon Press, 1930).
- ²⁷J. Burridge, "Statistical field theory for dialectology", *Physical Review E* **113**, 044310 (2026).
- ²⁸M. Davies, "Expanding horizons in historical linguistics with the 400-million word Corpus of Historical American English", *Corpora* **7**, 121–157 (2012).
- ²⁹A. Leemann, M.-J. Kolly, and D. Britain, "The English Dialects App: The creation of a crowdsourced dialect corpus", *Ampersand* **5**, 1–17 (2018).
- ³⁰B. Vaux and M. L. Jøhndal, *The Cambridge Online Survey of World Englishes*, https://www.tekstlab.uio.no/cambridge_survey/.
- ³¹J. Burridge and N. Tosh, *Inference in Statistical Modelling and Machine Learning: a concise introduction* (Cambridge University Press, 2026).

- ³²X. Guo, S. Verstyuk, H. Chen, B. Zhou, and S. Skiena, "Statistical structure and the evolution of languages", *Proceedings of the Royal Society B: Biological Sciences* **293**, 20252374 (2026).
- ³³J. Hofbauer and K. Sigmund, *Evolutionary Games and Population Dynamics* (Cambridge University Press, Cambridge, 1998).
- ³⁴C. E. Rasmussen and C. K. I. Williams, *Gaussian Processes for Machine Learning* (The MIT Press, 2006).
- ³⁵G. Bailey, T. A. Wikle, J. Tillery, and L. Sand, "The apparent time construct", *Language Variation and Change* **3**, 241–264 (1991).
- ³⁶G. Sankoff, "Cross-sectional and longitudinal studies in sociolinguistics", in *Sociolinguistics: An International Handbook of the Science of Language and Society, Vol. 2*, edited by U. Ammon, N. Dittmar, K. J. Mattheier, and P. Trudgill (Mouton de Gruyter, Berlin, 2005), pages 1003–1013.
- ³⁷G. Sankoff, "Language Change Across the Lifespan", *Annual review of linguistics* **4**, 297–316 (2018).
- ³⁸U.S. Census Bureau, *2020 Decennial Census*, United States Census Bureau, 2020.
- ³⁹T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd edition, Springer Series in Statistics (Springer, New York, 2009).
- ⁴⁰K. P. Murphy, *Probabilistic Machine Learning: An Introduction* (The MIT Press, 2022).
- ⁴¹J. Burridge and B. Vaux, *Supplemental Material for: Statistical Physics of Language Change Inferred from Time Evolving Maps*, Available with the published article, 2026.
- ⁴²G. K. Zipf, "The P1P2/D Hypothesis: On the Intercity Movement of Persons", *American Sociological Review* **11**, 677–686 (1946).
- ⁴³F. Simini, M. C. González, A. Maritan, and A.-L. Barabási, "A universal model for mobility and migration patterns", *Nature* **484**, 96–100 (2012).
- ⁴⁴Internal Revenue Service, *Statistics of Income: County-to-County Migration Data*, Internal Revenue Service, Statistics of Income Division, 2011.
- ⁴⁵U.S. Census Bureau, *Current Population Survey: Geographic Mobility/Migration Historical Tables*, U.S. Census Bureau, 2023.
- ⁴⁶U.S. Census Bureau, *CPS Historical Migration/Geographic Mobility Tables, Table A-1: Annual Geographic Mobility Rates, by Type of Movement: 1948–2023*, U.S. Census Bureau, Current Population Survey Annual Social and Economic Supplement, 2024.
- ⁴⁷W. Croft, *Explaining Language Change: An Evolutionary Approach* (Longman, London, 2000).
- ⁴⁸H. Giles, "Accent mobility: A model and some data", *Anthropological Linguistics* **15**, 87–109 (1973).
- ⁴⁹P. Trudgill, *New-Dialect Formation: The Inevitability of Colonial Englishes* (Edinburgh University Press, Edinburgh, 2004).
- ⁵⁰W. Labov, *Principles of Linguistic Change, Volume 2: Social Factors* (Blackwell, Oxford, 2001).
- ⁵¹X. Castelló, V. M. Eguíluz, and M. San Miguel, "Ordering dynamics with two non-excluding options: Bilingualism in language competition", *New Journal of Physics* **8**, 308 (2006).
- ⁵²U.S. Bureau of the Census, *Population of States and Counties of the United States: 1790 to 1990*, edited by R. L. Forstall (U.S. Government Printing Office, Washington, DC, 1996).
- ⁵³S. Kullback and R. A. Leibler, "On Information and Sufficiency", *Annals of Mathematical Statistics* **22**, 79–86 (1951).
- ⁵⁴W. Labov, "The social motivation of a sound change", *Word* **19**, 273–309 (1963).
- ⁵⁵H. Kurath, *A Word Geography of the Eastern United States* (University of Michigan Press, Ann Arbor, 1949).
- ⁵⁶W. Labov, *Principles of Linguistic Change, Volume 3: Cognitive and Cultural Factors* (Wiley–Blackwell, Oxford, 2010).
- ⁵⁷P. Kerswill, "Dialect Levelling and Geographical Diffusion in British English", in *Social Dialectology: In Honour of Peter Trudgill*, edited by D. Britain and J. Cheshire (Benjamins, Amsterdam, 2003), pages 223–243.
- ⁵⁸D. Britain, "Space and Spatial Diffusion", in *The Handbook of Language Variation and Change*, edited by J. K. Chambers, P. Trudgill, and N. Schilling-Estes (Blackwell, Oxford, 2002), pages 603–637.